Fault Code: 199 (Integrated)

ICON™ Lamp Circuit - Voltage Below Normal or Shorted to Low Source

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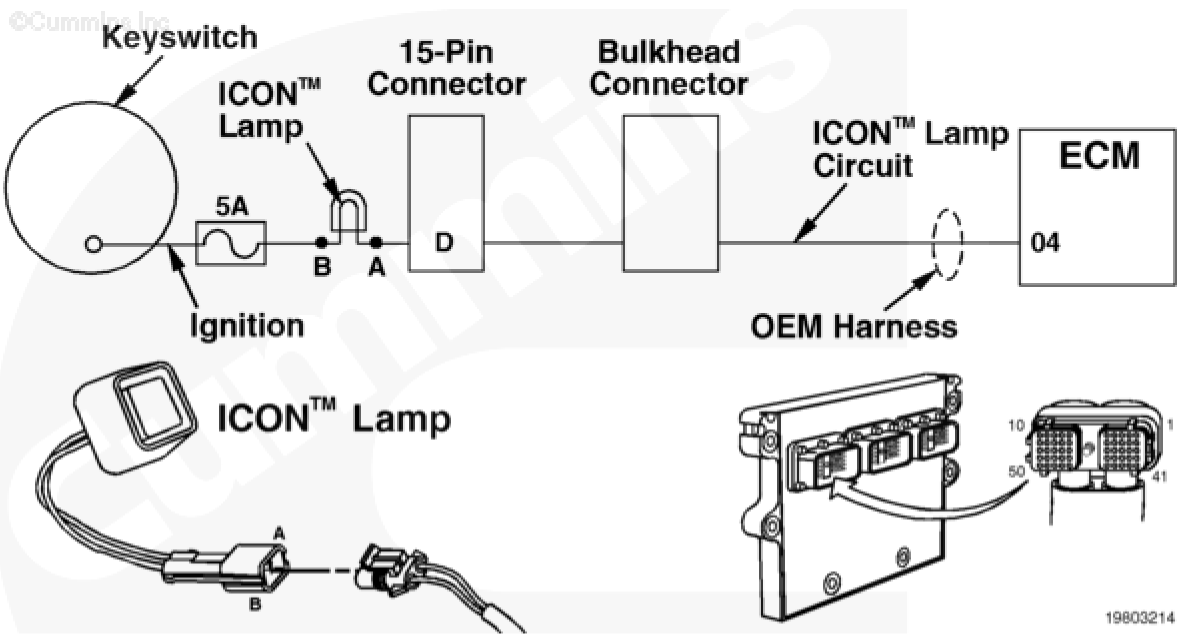
Overview

Codes	Reason	Effect
Fault Code: 199 PID(P): S122, 4 SPN: 612 FMI: 4 Lamp: Yellow SRT:	ICON™ Lamp Circuit - Voltage Below Normal or Shorted to Low Source. Less than 6 VDC (low voltage) detected at the ICON™ lamp circuit when high voltage was expected by the engine electronic control module (ECM).	Will **not** allow ICON™ to activate, however if ICON™ is engaged and fault code 199 becomes active, ICON™ will **not** be disabled.



Circuit Description

The ICON™ lamp circuit illuminates the ICON™ lamp to indicate when the ICON™ system is active. In addition, ICON™ active fault codes will be flashed out on this lamp. The lamp circuit requires a specific flash timing (on or off timing). If the on or off voltage is incorrect, the ICON™ system will be disabled. The lamp circuit **must** be functional to enable the ICON™ system.

Component Location

The ICON™ lamp is located in the vehicle cab on the dash panel.

Shoptalk

This fault indicates a short circuit to ground or an open circuit.

Warnings and Cautions

⚠ CAUTION ⚠

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement:
Part Number 3822917 - female Deutsch/AMP/Metri-Pack test lead
Part Number 3822758 - male Deutsch/AMP/Metri-Pack test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Read all fault codes.	
	STEP 1A. Read the fault codes with INSITE™ electronic service tool, or flash out engine lamp for active fault code.	Fault Code 199 inactive
STEP 2.	Check the ICON™ lamp.	
	STEP 2A. Check the bulb for continuity.	Less than 35 ohms
	STEP 2B. Check for voltage to the ICON™ lamp.	More than 6 VDC
STEP 3.	Check the fuse.	
	STEP 3A. Check the 5-amp ignition fuse.	Fuse installed correctly
	STEP 3B. Check if the 5-amp fuse is blown.	Fuse not blown

STEPS		SPECIFICATIONS
STEP 4.	Check the OEM cab harness connectors at the firewall and the OEM harness engine ECM connector.	
	STEP 4A. Inspect the OEM harness for damaged pins.	No damaged pins
STEP 5.	Check the ICON™ lamp circuit for an open or short circuit.	
	STEP 5A. Check the ICON™ lamp for an open circuit.	Less than 10 ohms
	STEP 5B. Check for a short circuit.	More than 100k ohms
	STEP 5C. Check for a short circuit from pin to pin.	More than 100k ohms
STEP 6.	Clear the fault code.	
	STEP 6A. Disable the fault code.	Fault Code 199 inactive

STEP 1. Read all fault codes.

STEP 1A. Read the fault codes with INSITE™ electronic service tool, or flash out engine lamp for active fault code.

Conditions :

- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Read the fault codes using INSITE™ electronic service tool, or flash out engine lamp for active fault codes.	Fault Code 199 inactive	**6A**
	Fault Code 199 active	**2A**

STEP 2. Check the ICON™ lamp.

STEP 2A. Check the bulb for continuity.

Conditions :

- Turn keyswitch OFF.
- Remove the bulb from the holder.

Action	Specification/Repair	Next Step
• Use a multimeter to check the resistance of the bulb. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Less than 35 ohms	**2B**
	Replace the bulb for a 12-VDC system with a General Electric 1892 bulb or equivalent.	**6A**

STEP 2B. Check for voltage to the ICON™ lamp.

Conditions :

- Remove the bulb from the holder.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step
• Use a multimeter to measure the voltage at the lamp bulb holder keyswitch side (power side) to chassis ground. For multimeter usage techniques, refer to Multimeter Usage, Procedure 019-359 (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html).	More than 6 VDC	**4A**
	Less than 6 VDC	**3A**

STEP 3. Check the fuse.

STEP 3A. Check the 5-amp ignition fuse.

Conditions :

- Turn keyswitch OFF.

Action	**Specification/Repair**	**Next Step**
Refer to Procedure 019-198 (/qs3/pubsys2/xml/en/procedures/99/99-019-198.html).	Fuse installed correctly	**3B**
	Install fuse correctly. Refer to Procedure 019-198 (/qs3/pubsys2/xml/en/procedures/99/99-019-198.html).	**6A**

STEP 3B. Check if the 5-amp fuse is blown.

Conditions :

- Turn keyswitch OFF.

Action	**Specification/Repair**	**Next Step**
Refer to Procedure 019-198 (/qs3/pubsys2/xml/en/procedures/99/99-019-198.html).	Fuse not blown	**4A**
	Replace the blown fuse. Refer to Procedure 019-198 (/qs3/pubsys2/xml/en/procedures/99/99-019-198.html).	**6A**

STEP 4. Check the OEM cab harness connectors at the fire wall and the OEM harness engine ECM connector.

STEP 4A. Inspect the OEM harness for damaged pins.

Conditions :

- Turn keyswitch OFF.

Action	**Specification/Repair**	**Next Step**

• Corroded pins • Bent or broken pins • Pushed back or expanded pins • Wire insulation damage • Moisture in or on the connector • Missing or damaged connector seals • Connector shell broken • Dirt or debris in or on the connector pins. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	**5A**
	Repair or replace the OEM harness. Refer to the OEM service manual.	**6A**

STEP 5. Check the ICON™ lamp circuit for an open or short circuit.

STEP 5A. Check the ICON™ lamp for an open circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness connector from the engine ECM.
- Remove the ICON™ lamp bulb from the bulb holder.

Action	Specification/Repair	Next Step

• Measure the resistance from pin 4 of the OEM harness connector at the engine ECM to the return side (engine ECM side) of the ICON™ lamp bulb holder. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Less than 10 ohms	**5B**
	Repair or replace the OEM harness. Refer to the OEM service manual.	**6A**

STEP 5B. Check for a short circuit.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness connector from the engine ECM.
- Remove the ICON™ lamp bulb from the bulb holder.

Action	Specification/Repair	Next Step

• Measure the resistance from pin 4 of the OEM harness connector at the engine ECM to ground.	More than 100k ohms	**5C**
Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Repair or replace the OEM harness. Refer to the OEM service manual.	**6A**

STEP 5C. Check for a short circuit from pin to pin.

Conditions :

- Turn keyswitch OFF.
- Disconnect the OEM harness connector from the engine ECM.
- Remove the ICON™ lamp bulb from the bulb holder.

Action	Specification/Repair	Next Step

• Measure the resistance from pin 4 of the OEM harness connector at the engine ECM to all other pins in the connector. Refer to the wiring diagram or the circuit description at the beginning of this fault code for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 100k ohms	**6A**
	Repair or replace the OEM harness. Refer to the OEM service manual.	**6A**

STEP 6. Clear the fault code.

STEP 6A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step

	Fault Code 199 inactive	**Repair complete**
- Verify that Fault Code 199 is inactive. - Erase the inactive fault codes using INSITE™.	Return to the troubleshooting steps, or contact the nearest Cummins Authorized Repair Location if all the steps have been completed and rechecked. Troubleshoot any remaining active fault codes.	**Appropriate troubleshooting charts**

Last Modified: 04-Oct-2004